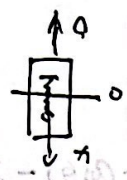


1.



$$kx - mg = ma$$

$$\therefore x = \frac{m(g+a)}{k}$$

$$-mgx + \frac{1}{2}kx^2 = \frac{1}{2}kx_0^2 - mgx_0 + \frac{1}{2}mv^2$$

E_{max}

$$E_{\text{max}} = \frac{m^2 g^2}{2k}$$

2.

$$\vec{r} = a \cos(\omega t) \vec{i} + b \sin(\omega t) \vec{j}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = -a\omega \sin(\omega t) \vec{i} + b\omega \cos(\omega t) \vec{j}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = -a\omega^2 \cos(\omega t) \vec{i} - b\omega^2 \sin(\omega t) \vec{j}$$

$$\vec{F} = m\vec{a} = -ma\omega^2 \cos(\omega t) \vec{i} - mb\omega^2 \sin(\omega t) \vec{j}$$

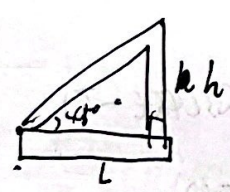
$$\vec{M} = \vec{r} \times \vec{F} = (a \cos(\omega t) \vec{i} + b \sin(\omega t) \vec{j}) \times (-ma\omega^2 \cos(\omega t) \vec{i} - mb\omega^2 \sin(\omega t) \vec{j}) = -mab\omega^2 \sin(\omega t) \cos(\omega t) \vec{k}$$

$$\vec{L} = \vec{r} \times \vec{p} = (a \cos(\omega t) \vec{i} + b \sin(\omega t) \vec{j}) \times m(-a\omega \sin(\omega t) \vec{i} + b\omega \cos(\omega t) \vec{j}) = 0$$

$$= -mab\omega^2 \sin(\omega t) \cos(\omega t) \vec{k} + mab\omega^2 \sin(\omega t) \cos(\omega t) \vec{k}$$

$$= mab\omega^2 \vec{k}$$

3.



设, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

$$L = l \sqrt{1 - \beta^2}, \quad \beta = \left(\frac{v}{c}\right)$$

$$\therefore L = l \sqrt{1 - \beta^2}$$

$$L \sqrt{1 - \left(\frac{v}{c}\right)^2} = h$$

$$\sin \theta = \frac{h}{l}$$

$$L = \sqrt{3} h$$

$$L \sqrt{1 - \left(\frac{v}{c}\right)^2} = \sqrt{3} h \sqrt{1 - \left(\frac{v}{c}\right)^2} \quad v = \sqrt{3} h^2 / c^2$$



4. 碰

5.

6.

(1) 碰撞前后 (碰撞前后) 水平动量守恒

$$L_0 = \frac{2}{3}L \cdot mv$$

$$L_0 = -L_0 + J\omega$$

$$J = J_m + J_{lm} + J$$

$$= m \cdot \left(\frac{2}{3}v\right)^2 + 2m \cdot \left(\frac{1}{3}v\right)^2 + \frac{1}{12} \cdot 3mL^2 + 3m \cdot \left(\frac{1}{6}v\right)^2$$

$$= mL^2$$

$$\text{角速度 } \omega = \frac{4v}{3L}$$

(2) 由于碰撞前后角动量守恒

$$\therefore v' + \omega \frac{2}{3}L = v$$

$$\therefore v' = v - \omega \frac{2}{3}L$$

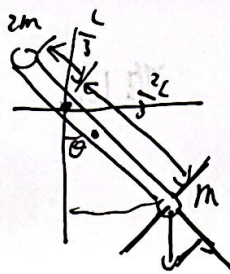
$$\therefore L_0 = L' + J\omega$$

$$mv \cdot \frac{2}{3}L = -m(v - \omega \frac{2}{3}L) \cdot \frac{2}{3}L + mL^2 \cdot \omega$$

$$\omega = \frac{2v}{3L}$$

(2)

(3)

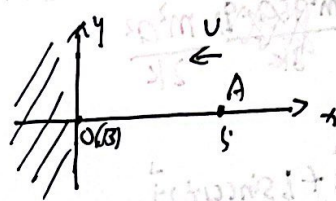


$$\sum J\omega^2 = mg \frac{2L}{3} \cdot (1 - \cos\theta) - 2mg \cdot \frac{L}{3} (1 - \cos\theta) + 3mg \cdot \frac{L}{6} (1 - \cos\theta)$$

$$\theta = \arccos\left(\frac{9 - \omega^2 L^2}{9}\right)$$

7.

(1)



$$\text{设 } y = A \cos(\omega t + kx + \phi)$$

$$\omega = 4\pi, A = 3$$

$$k = \frac{2\pi}{\lambda} = \frac{\omega}{v} = \frac{\pi}{5}$$

又: A 在 B 5m

$$\therefore 4\pi t + \frac{\pi}{5} \cdot 5 + \phi = 4\pi t$$

$$y_1 = 3 \cos(4\pi t + \frac{\pi}{5}x - \pi)$$

(2) 由波速为 v 及波长为 \lambda

故有半波损失

$$\text{入射波在 D 处的振动为 } y_1 = 3 \cos(4\pi t - \pi)$$

$$\therefore \text{反射波振动为 } y_2 = 3 \cos(4\pi t)$$

$$\therefore \text{反射波波函数为 } y_2 = 3 \cos(4\pi t - \frac{\pi}{5}x)$$

$$\text{合波 } y = y_1 + y_2$$

$$= 2 \sin(4\pi t) \sin(\frac{\pi}{5}x)$$

$$\text{当 } \sin(\frac{\pi}{5}x) = 0 \text{ 时}$$

$$x = 5k$$

故 A、B 均为波节

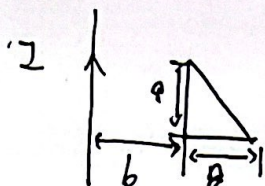


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8.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

$$\therefore B = \frac{\mu_0 I}{2\pi r}$$



$$\begin{aligned} \Phi &= \int \vec{B} \cdot d\vec{S} \\ &= \int_b^{b+a} \frac{\mu_0 I}{2\pi r} \cdot a \cdot dr \\ &= \int_b^{b+a} B(a+b-x) dx \\ &= \int_b^{b+a} \frac{\mu_0 I}{2\pi x} (a+b-x) dx \\ &= \frac{\mu_0 I}{2\pi} \left[(a+b) \ln \frac{a+b}{b} - a \right] \\ &\quad \text{取 } a=0 \text{ 时 } \Phi(0) = 0 \\ &\therefore \Phi(0) = 0 \end{aligned}$$

(2) 由洛伦兹力方向可知

故 ~~b(t) = b + vt~~

$$b = b + vt$$

$$\therefore \Phi(t) = \frac{\mu_0 I}{2\pi} \left[(a+b+vt) \ln \frac{a+b+vt}{b+vt} - a \right]$$

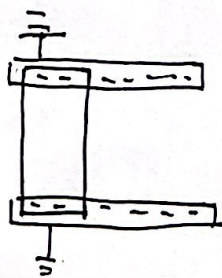
$$\mathcal{E} = - \frac{d\Phi(t)}{dt}$$

$$\begin{aligned} &= - \frac{\mu_0 I}{2\pi} \left\{ \ln \cos(\omega t) \left[(a+b+vt) \ln \frac{a+b+vt}{b+vt} - a \right] \right. \\ &\quad \left. + \sin \omega t \cdot \left[v \ln \frac{a+b+vt}{b+vt} - \frac{v a}{b+vt} \right] \right\} \end{aligned}$$

同轴电缆系统
由静电平衡可知

$$\oint \vec{E} \cdot d\vec{S} = \frac{\Sigma Q}{\epsilon_0}$$

设内上电荷在电荷分布为 σ ($\sigma < 0$)



$$\therefore \oint \vec{E} \cdot d\vec{S} = \frac{\Sigma Q}{\epsilon_0}$$

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$$\therefore \oint \vec{E} \cdot d\vec{S} = \frac{\Sigma Q}{\epsilon_0}$$

$$\therefore \sigma = n d q$$

$$\therefore \oint \vec{E} \cdot d\vec{S} = \frac{\Sigma Q}{\epsilon_0}$$

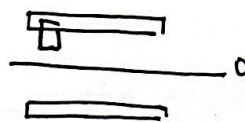
~~取~~

$$\oint \vec{E} \cdot d\vec{S} = \frac{\Sigma Q}{\epsilon_0}$$

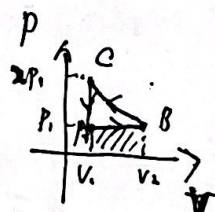
$$E = (d-d') \frac{n q}{\epsilon_0} = \frac{n q x}{\epsilon_0}$$

$$U = \int_0^x E dx$$

$$\begin{aligned} &= \int_0^x \frac{n q x}{\epsilon_0} dx \\ &= \frac{n q (d^2 - x^2)}{2 \epsilon_0} \end{aligned}$$



10.



(1) A $\rightarrow$ B, P 不变

设初末态温度为 T

$$P_1 V_1 = \frac{m}{M} R T_1$$

$$T_1 = \frac{P_1 V_1}{R}$$

$$\therefore \frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$T_2 = 2 T_1$$

$$\therefore V_2 = 2 V_1$$

$$Q_1 = \frac{m}{M} C_p \Delta T$$

$$C_p = \frac{5}{2} R = \frac{7}{2} R$$

$$\therefore Q_1 = \frac{7}{2} P_1 V_1$$

$$W_1 = \int_{V_1}^{V_2} P_1 dV$$

$$= P_1 V_1$$

B $\rightarrow$ C 等温 $\Delta E_2 = 0$

$$Q_2 = A W_2$$

$$W_2 = \int_{V_1}^{V_2} P_2 dV$$

$$\text{且 } P V = P_2 V_2$$

$$\therefore W_2 = -2 \ln 2 P_1 V_1$$

$$Q_2 = -2 \ln 2 P_1 V_1$$

C $\rightarrow$ A 等压 $W_3 = 0$

$$Q_3 = \Delta E_3$$

$$= \frac{m}{M} C_p \Delta T$$

$$= -\frac{5}{2} P_1 V_1$$

$$W = W_1 + W_2 + W_3 = P_1 V_1 (1 - \ln 4) \quad \therefore \eta = \frac{1 - \ln 4}{1}$$

$$Q_2 = Q_1 + Q_3$$

$$= (1 - \ln 4) P_1 V_1$$



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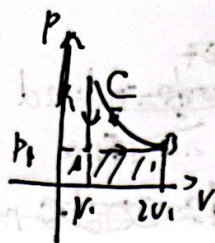
(1) 求变

由于A→B为可逆过程

$$\Delta S = \int_A^B \frac{dQ}{T}$$

$$dQ = G \cdot dT$$

$$\Delta S = \frac{G}{n \cdot R} \ln \frac{T_2}{T_1}$$



$$2.5 \times 10^{-2} \times 10^5 = 2.5 \times 10^3$$

$$2.5 \times 10^3 = 2.5 \times 10^3$$

$$\frac{2.5}{2} = 1.25 \times 10^3$$

$$\frac{2.5 \times 10^3}{2} = 1.25 \times 10^3$$

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$$P_1 V_1 = P_2 V_2$$

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